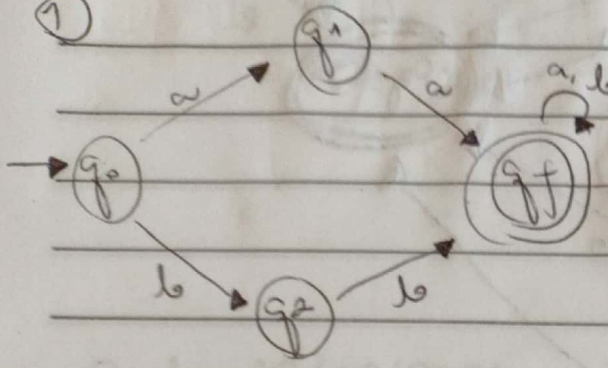
Exercícios Assíncronos (Conversão AFND para AFD):

Seja os seguintes AFND's, encontrar o respectivo AFD:

1)



δ_1	a	b
→ q_0	$\{q_1\}$	$\{q_2\}$
q_1	$\{q_f\}$	–
q_2	–	$\{q_f\}$
* q_f	$\{q_f\}$	$\{q_f\}$

$L1 = \{w \in \{a,b\}^* \mid$ "aa" ou "bb" é prefixo de $w\}$

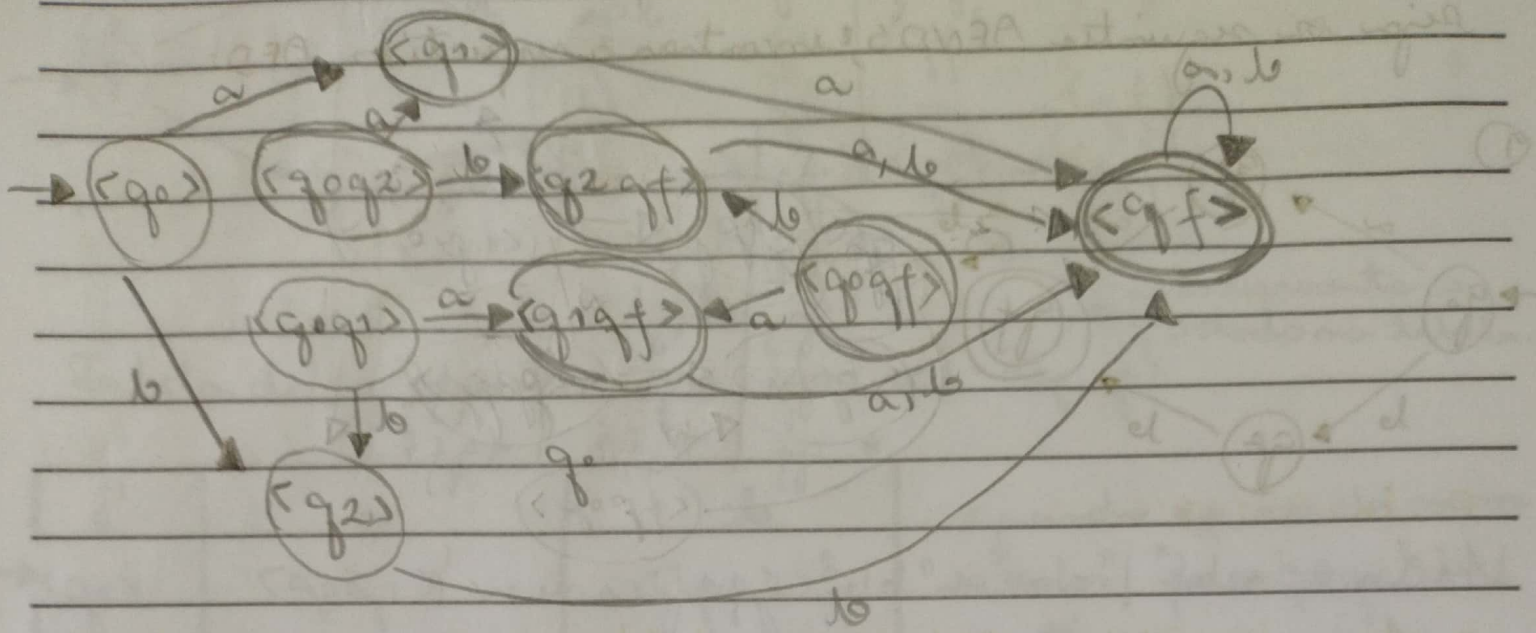
Tabela de conversão:

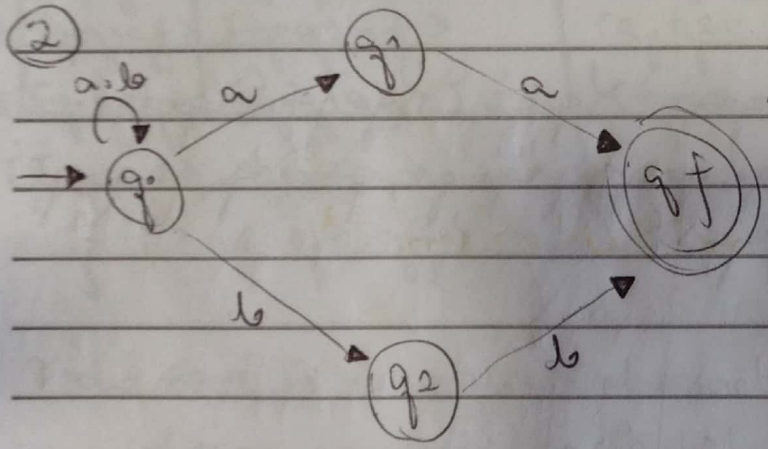
$2^n = \{\{q_0\}, \{q_1\}, \{q_2\}, \{q_f\}, \{q_0q_1\}, \{q_0q_2\}, \{q_0q_f\}, \{q_1q_2\}, \{q_1q_f\}, \{q_2q_f\}, \{q_0,q_1,q_2\}, \{q_0,q_1,q_f\}, \{q_0,q_2,q_f\}, \{q_0,q_1,q_2,q_f\}\}$

$Q' = \{\langle q_0\rangle, \langle q_1\rangle, \langle q_2\rangle, \ldots, \langle q_0q_1q_2q_f\rangle\}$

δ'	a	b
→ $\langle q_0\rangle$	$\langle q_1\rangle$	$\langle q_2\rangle$
$\langle q_1\rangle$	$\langle q_f\rangle$	–
$\langle q_2\rangle$	–	$\langle q_f\rangle$
* $\langle q_f\rangle$	$\langle q_f\rangle$	$\langle q_f\rangle$
$\langle q_0q_1\rangle$	$\langle q_1q_f\rangle$	$\langle q_2\rangle$
$\langle q_0q_2\rangle$	$\langle q_1\rangle$	$\langle q_2q_f\rangle$
* $\langle q_0q_f\rangle$	$\langle q_1q_f\rangle$	$\langle q_2q_f\rangle$
$\langle q_1q_2\rangle$	$\langle q_f\rangle$	$\langle q_f\rangle$
* $\langle q_1q_f\rangle$	$\langle q_f\rangle$	$\langle q_f\rangle$
* $\langle q_2q_f\rangle$	$\langle q_f\rangle$	$\langle q_f\rangle$

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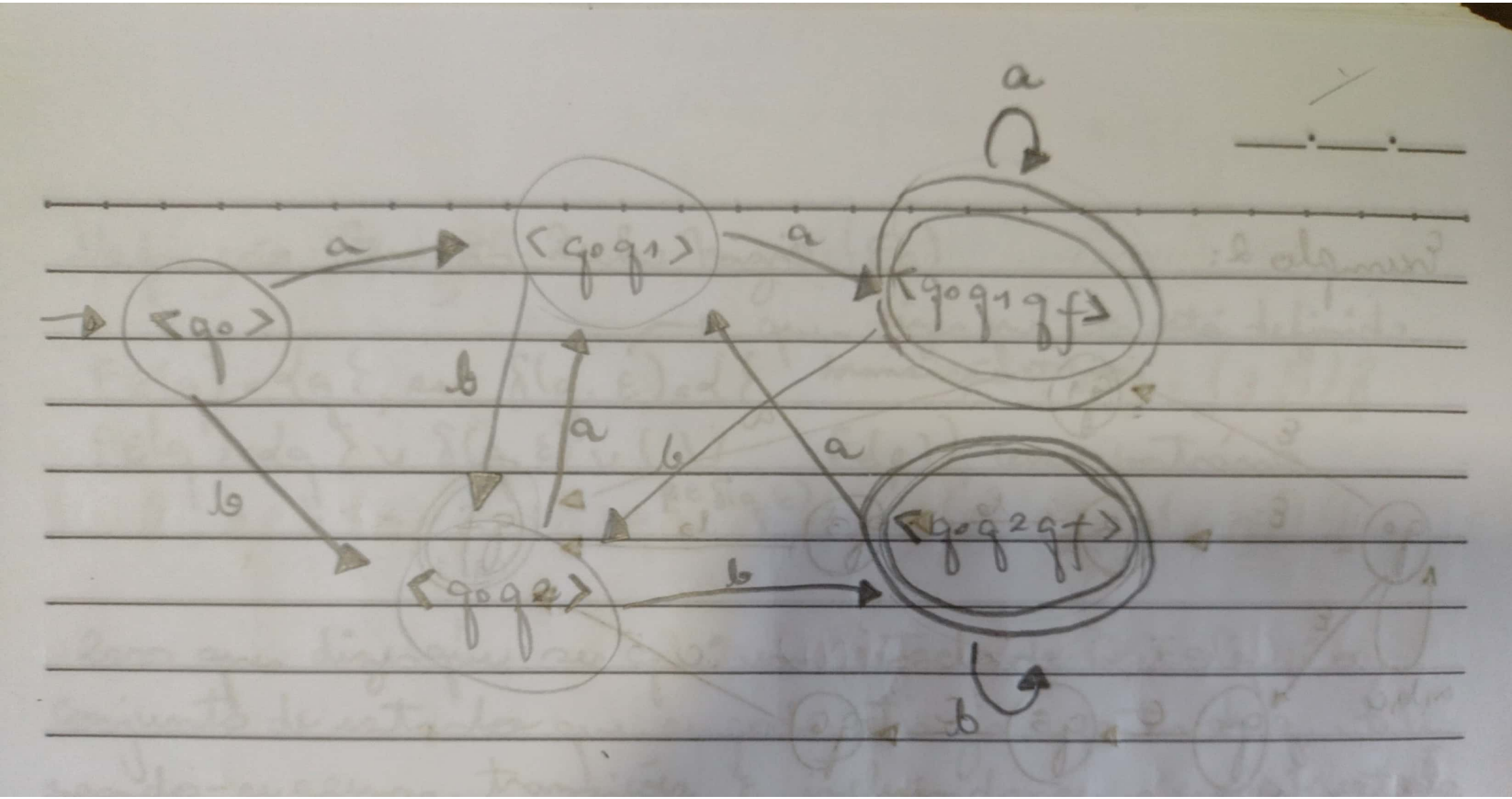


δ_2	a	b
→ q0	{q0, q1}	{q0, q2}
q1	{qf}	—
q2	—	{qf}
*qf	—	—

$L2 = \{ w \in \{a, b\}^* \mid$ "aa" ou "bb" é sufixo de $w \}$

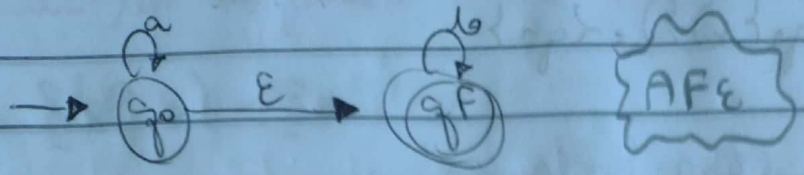
δ'	a	b
→ ⟨q0⟩	⟨q0q1⟩	⟨q0q2⟩
⟨q0q1⟩	⟨q0q1qf⟩	⟨q0q2⟩
~~⟨q0q2⟩~~	⟨q0q1⟩	⟨q0q2qf⟩
* ~~⟨q0q1qf⟩~~	⟨q0q1qf⟩	⟨q0q2⟩
* ~~⟨q0q2qf⟩~~	⟨q0q1⟩	⟨q0q2qf⟩

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Exercício Assíncrono:



• $M'_{10} = (\{a,b\}, \{q_0, q_f\}, \delta_{10}, q_0, F'\}$

$F\varepsilon\{q_0\} = \{q_0, q_F\}$

$F\varepsilon\{q_f\} = \{q_F\}$

$F' = \{q_0, q_F\}$ → novos estados de aceitação

• $\delta'_{10}(q_0, a) = \underline{\delta}_{10}(\{q_0\}, a) =$ q_0 lendo a = OK; q_F lendo a = indefinido

$F\varepsilon(\{r_n / r_n \in \delta(s,a) \text{ e } s \in \underline{\delta}(\{q_0\}, \varepsilon)\}) = F\varepsilon(\{q_0\}) = \{q_0, q_F\}$

$\{q_0, q_F\}$

• $\delta'_{10}(q_0, b) = \underline{\delta}_{10}(\{q_0\}, b) =$ q_0 lendo b indefinido; q_F lendo b = OK!

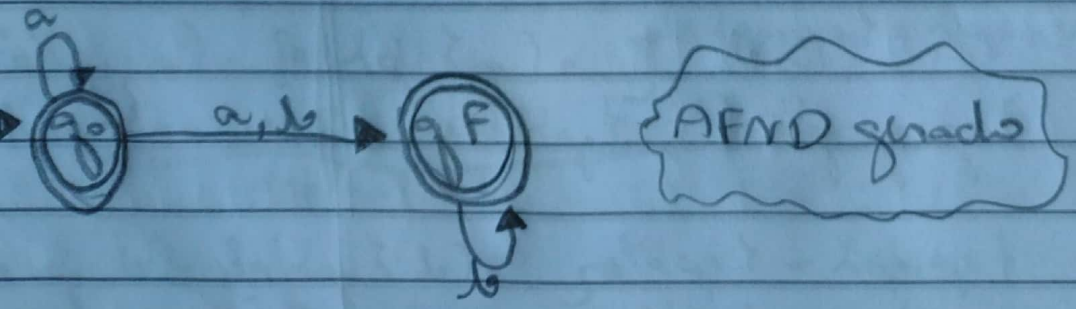
$F\varepsilon(\{r_n / r_n \in \delta(s,b) \text{ e } s \in \underline{\delta}(\{q_0\}, \varepsilon)\}) = F\varepsilon(\{q_F\}) = \{q_F\}$ ←

• $\delta'_{10}(q_F, a) = \underline{\delta}'_{10}(\{q_F\}, a) =$ q_F lendo a →

$F\varepsilon(\{r_n / r_n \in \delta(s,a) \text{ e } s \in \underline{\delta}(\{q_F\}, \varepsilon)\}) = F\varepsilon(\{\}) =$ indefinido

• $\delta'_{10}(q_F, b) = \underline{\delta}'_{10}(\{q_F\}, b) =$

$F\varepsilon(\{r_n / r_n \in \delta(s,b) \text{ e } s \in \underline{\delta}(\{q_F\}, \varepsilon)\}) = F\varepsilon(\{q_F\}) = \{q_F\}$



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